LESSON PLAN TEMPLATE

PREPARE

Objectives: By the end of the lesson the students will be able to… (how will you know they know?)

1. Students will be able to compute expected values and standard deviations of sums and averages of random variables.
2. Students will be able to apply Central Limit Theorem and Law of Large Number in examples to make predictions and assess their predictions.
3. Students will be able to set up probability equations for some real applications and solve the equations by normal approximation.

Materials:

- Tablet, handout with notes (attached)

Personal goals (what I'm working on in my teaching):

1. Clarity of explaining steps while writing
2. Interactions with students, including facing to students, initiating activities, asking for feedbacks, answering questions, giving feedbacks to students' answers.
3. Effective use of tablet (this is the first time I use a tablet to teach/give a presentation!)
4. I also want to practice time management, but as I discussed with Tori, it is likely that we cannot cover everything we prepared — let's see :)

INSTRUCT

Intro:

Brief Self-introduction:

- Name, undergrad institution, current area of interest (connection to probability)
- Thank for the opportunity of practice teaching

Bridge-in to create interest and motivation to learn:

- Tori has presented the concept/statement of Central Limit Theorem. In the rest of the class, we will focus on practicing this principle in example. We will see the power of Central Limit Theorem in real-world problems that allow us to compute probability with very limited information.

State learning objectives and connections:

- Compute means and standard deviations of sums and averages of random variables
  - Practice properties of expected value and standard deviation
- Apply Central Limit Theorem to a big example
  - Practice setting up and solving probability equations using normal approximation

Learning Activities: What you will do, what they will do, check for understanding (assess). Connect activities to each other. Repeat.

Activity: Compute means and standard deviations of sums and averages of random variables (~15min)

- Students will work on the computations themselves for ~2min and discuss with their friends for ~2min

Assess: I will invite students to give their answers:

- If their answer is correct, I will record and present their steps to class

- If not, we will work through their solution together and see what went wrong. We will work out the adjustment accordingly

Connect: Remind students that we are now ready to apply Central Limit Theorem in practice!

Activity: Fibonacci number example - introduction and predict the mean of average (~5min)
Assess: Invite answers/guesses of the prediction with guidance:
- What are we looking for?
- What's a good guess? Why?

Connect: *Central Limit Theorem and Law of Large Number*

Activity: Fibonacci number example - Problem 2 (~11min)
- Students will work in groups on parsing the problem and setting up the probability equation (with a hint of what a probability equation looks like) ~3min

Assess: I will invite students to give answers and work though the setup together. I will share my idea/trick of parsing the problem. Then, I will present the complete steps of solving the equation.
Connect: Application of normal approximation

Activity: Fibonacci number example - Problem 3 (~6min)
- Students will work in groups on parsing the problem and setting up the probability equation ~3min

Assess: I will invite students to give answers and work though the setup.
- If we have >15min left, then I will ask students to solve the equation. Most likely otherwise , I will mention that it's the same as the last problem and encourage students to practice in their own time

Connect: Application of normal approximation

Activity: Fibonacci number example - Problem 4 (~8min)
- Students will work in groups on parsing the problem and setting up the probability equation ~3min

Assess: I will invite students to give answers. If time permits, I will ask students to solve the equation. Otherwise, we will walk through the steps together (or leave as an exercise).
Connect: Central Limit Theorem and application of normal approximation

Closing:

Summarize lesson:
- Recap of solving problems with Central Limit Theorem.

Mean and standard deviation

Suppose $X_1, \ldots, X_n$ are independent, all with the same distribution. $\mathbb{E}(X_i) = \mu$ and $SD(X_i) = \sigma$.

1. What is $\mathbb{E}(X_1 + X_2 + \cdots + X_n)$?

$$\mathbb{E}(X_1 + X_2 + \cdots + X_n)$$
$$= \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_n)$$
$$= \mu + \mu + \cdots + \mu$$
$$= n\mu$$

2. What is $SD(X_1 + X_2 + \cdots + X_n)$?

$$SD(X_1 + X_2 + \cdots + X_n).$$
$$= \sqrt{Var(X_1 + X_2 + \cdots + X_n)}$$
$$= \sqrt{Var(X_1) + Var(X_2) \cdots + Var(X_n)}$$
$$= \sqrt{\sigma^2 + \sigma^2 + \cdots + \sigma^2}$$
$$= \sqrt{n}\sigma$$

& $SD(X) = \sigma$
$\Rightarrow Var(X) = \sigma^2$

3. What is $\mathbb{E}\left(\dfrac{X_1 + X_2 + \cdots + X_n}{n}\right)$?

$$\mathbb{E}\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right)$$
$$= \mathbb{E}\left(\frac{X_1}{n}\right) + \mathbb{E}\left(\frac{X_2}{n}\right) + \cdots + \mathbb{E}\left(\frac{X_n}{n}\right)$$
$$= \frac{1}{n}\mathbb{E}(X_1) + \frac{1}{n}\mathbb{E}(X_2) + \cdots + \frac{1}{n}\mathbb{E}(X_n)$$
$$= \frac{1}{n}(n\mu)$$
$$= \mu.$$

4. What is $SD\left(\dfrac{X_1 + X_2 + \cdots + X_n}{n}\right)$?

$$SD\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right)$$
$$= \sqrt{Var\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right)}$$
$$= \sqrt{Var\left(\frac{X_1}{n}\right) + Var\left(\frac{X_2}{n}\right) + \cdots Var\left(\frac{X_n}{n}\right)}$$
$$= \sqrt{\frac{\sigma^2}{n^2} + \frac{\sigma^2}{n^2} + \cdots + \frac{\sigma^2}{n^2}}$$
$$= \sqrt{\frac{\sigma^2}{n}}$$
$$= \frac{\sigma}{\sqrt{n}}.$$

Hints for all problems:

Additivity of $\mathbb{E}$:

$\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$

Additivity of Var:

$Var(X+Y) = Var(X) + Var(Y)$ if X, Y independent

$\mathbb{E}(nX) = n\mathbb{E}(X)$

$Var(nX) = n^2 Var(X)$

! $SD(X_1 + \cdots + X_n) \neq SD(nX) = n\sigma.$

since $\underbrace{X_1 \cdots X_n}_{n \text{ trials}}$ are independent

$nX = 1 \text{ trial} \times n.$

Agenda

① Introduction

② Mean, SD of sums + avgs

* use for N.A.

③ Big example

Structure of this topic

① Repeat the motivation: use of N.A. for sum & avg.

② ~2min work individually thru the problems

③ ~2min check in with friends

④ Ask for students' responses

→ correct: record the steps

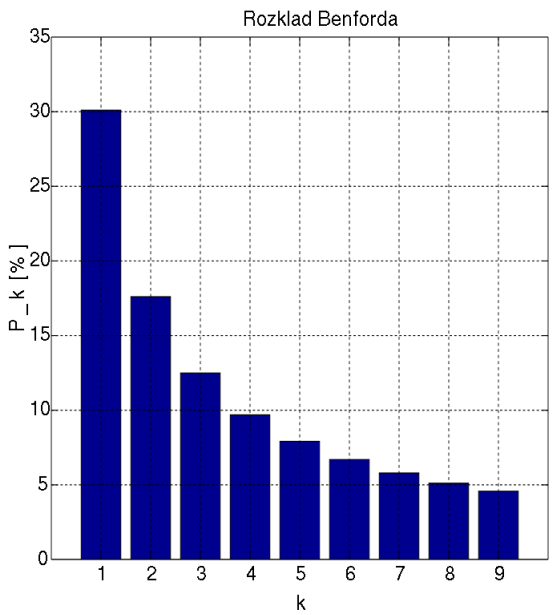
→ incorrect: work thru/try to understand what went wrong and present the correct way.

! Any questions?

Example: Law of Averages and Central Limit Theorem

Let D_i be the first digit of a randomly chosen Fibonacci number (from among the first 100,00 Fibonacci numbers). (Benford's Law!)

1, 1, 2, 3, 5, 8 ...



Apply C.L.T. and formulae above to an example

$\mathbb{E}(D_i) = 3.44$

$SD(D_i) = 2.4611$

Define $\bar{D}_n$ as the *average* of n randomly selected Fibonnacci numbers:

$$\bar{D}_n = \frac{D_1 + \cdots + D_n}{n} \tag{1}$$

1. Predict the value of $\bar{D}_n$ when n is large.

Explain and ask for responses: ① what are we looking for? ② what's a good guess? why?

Law of Large number: $\mathbb{E}(\bar{D}_n) = \mathbb{E}(D_i) = 3.44$.

2. Find the number ϵ such that for $n = 10,000$ the chance that your prediction is off by more than ϵ is about 10%.

$\mathbb{P}(|\bar{D}_n - 3.44| > \varepsilon) = 0.1$

① $1 - \mathbb{P}(|\bar{D}_n - 3.44| \le \varepsilon) = 0.1$

$1 - \mathbb{P}(-\varepsilon \le \bar{D}_n - 3.44 \le \varepsilon) = 0.1$

$\mathbb{P}(-\varepsilon \le \bar{D}_n - 3.44 \le \varepsilon) = 0.9$

② $\mathbb{P}(-\varepsilon + 3.44 \le \bar{D}_n \le \varepsilon + 3.44) = 0.9$

③ $\mathbb{P}\left(\frac{-\varepsilon + 3.44 - \mu}{\frac{\sigma}{\sqrt{n}}} \le \frac{\bar{D}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \le \frac{\varepsilon + 3.44 - \mu}{\frac{\sigma}{\sqrt{n}}}\right) = 0.9.$

$\mathbb{E}(\bar{D}_n) = \mathbb{E}(D_i) = \mu.$

$SD(\bar{D}_n) = \frac{SD(D_i)}{\sqrt{n}} = \frac{\sigma}{\sqrt{n}}$

Structure of this problem

① ~3min Activity: Parse the problem and set up a prob. eqn: $\mathbb{P}(\quad) = (\quad)$

② Ask for responses.

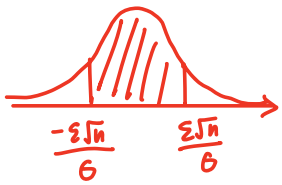
③ Solve the eqn together!

i). get rid of $|\cdot|$

ii) separate variable

iii) standize the dist.

iv) rewrite using Φ



$$\mathbb{P}\left(\frac{-\varepsilon\sqrt{n}}{\sigma} \le X \le \frac{\varepsilon\sqrt{n}}{\sigma}\right) = 0.9$$

④ $$\Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right) - \Phi\left(\frac{-\varepsilon\sqrt{n}}{\sigma}\right) = 0.9$$

$$\Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right) - \left(1 - \Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right)\right) = 0.9$$

$$2\,\Phi\left(\frac{\varepsilon\sqrt{n}}{\sigma}\right) = 1.9$$

$$\Phi\left(\frac{\varepsilon\sqrt{n}}{2.4611}\right) = 0.95.$$

⑤ $$\frac{\varepsilon\sqrt{10000}}{2.4611} = \Phi^{-1}(0.95)$$

⑥ look up the table:

$$\frac{100\varepsilon}{2.4611} = 1.645.$$

$$\varepsilon \approx 0.0405.$$

v) plug in σ

vi) look up the table!

vii) Solve with n.

④ Questions?

3. Find approximately the least value of n such that your prediction of $\bar{D}_n$ is correct to within 0.01 with probability at least 0.95.

$\mathbb{P}(|\bar{D}_n - 3.44| \le 0.01) \ge .95$

① $\mathbb{P}(-0.01 \le \bar{D}_n - 3.44 \le 0.01) \ge .95$

② $\mathbb{P}\left(\frac{-0.01\sqrt{n}}{6} \le \bar{D}_n \le \frac{0.01\sqrt{n}}{6}\right) \ge .95$

③ $2\Phi\left(\frac{0.01\sqrt{n}}{6}\right) - 1 \ge .95$

$\Phi\left(\frac{0.01\sqrt{n}}{2.4611}\right) \ge 0.975$

④ $\frac{0.01\sqrt{n}}{2.4611} \ge \Phi^{-1}(0.975) = 1.96$

⑤ $\sqrt{n} \ge \frac{1.96 \times 2.4611}{0.01} = 482.3756$

⑥ $n \ge 232686.219$

$n = 2326867.$

Structure of this problem

① ~3min activity: parse the problem and set up the probability eqn.

② Ask for answers.

③ - if > 15 min, then ask students to solve the eqn ~3min. and go thru the solution ~2min.

- otherwise, say that the solving process is similar as above and encourage students to solve it in their time for practice.

4. If you had to predict the first digit of $\bar{D}_{100}$, what digit should you choose to maximize your chance of being correct, and what is that chance?

Choice of prediction: 3, since $\mathbb{E}(\bar{D}_n) = 3.44$.

Chance: all numbers start with 3 count, so:

$\mathbb{P}(3 \le \bar{D}_n < 4)$

① $= \mathbb{P}\left(\frac{(3-3.34)\sqrt{n}}{6} \le \bar{D}_n < \frac{(4-3.34)\sqrt{n}}{6}\right)$

$= \mathbb{P}\left(\frac{3.4}{2.4611} \le \bar{D}_n < \frac{13.4}{2.4611}\right)$

$= \mathbb{P}(1.3815 \le \bar{D}_n < 5.4447)$

② $= \Phi(5.4447) - \Phi(1.3815)$

③ $= 1 - 0.9162$

$= 0.0838$

Structure of this problem

① ~3min. Activity: understand the problem and set up the probability eqn.

② Ask for answers.

③ ~3min Ask to try to solve the eqn.

④ Ask for solutions and go thru the steps togethers.

Questions or comments?